

PALMA DE MALLORCA AP, Spain

WMO: 083060

Lat: 39.561N

Lon: 2.737E

Elev: 26

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.2	35.4	28.8	22.8	41.1	31.6	25.9	41.0	27.3	56.4	24.3	56.0	1.7	40	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.4	92.7	72.2	89.8	72.5	87.7	72.4	77.9	85.5	76.5	84.3	75.3	83.4	10.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.5	133.5	82.4	73.8	126.2	81.6	72.8	121.7	81.0	41.3	85.4	39.9	84.6	38.8	83.3	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	22.9	19.9	17.6	DB	28.6	99.1	1.8	2.7	27.3	101.1	26.2	102.7	25.2	104.2	23.9	106.2
(5)				WB	28.0	80.8	1.7	2.3	26.8	82.5	25.8	83.8	24.8	85.1	23.6	86.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.8	50.2	50.1	53.4	58.1	64.6	72.3	77.3	78.0	72.2	66.2	57.3	52.3
	DBStd	10.91	4.42	4.58	4.35	3.95	4.36	4.57	3.55	3.44	4.24	4.48	5.12	4.64
	HDD50	152	51	49	17	1	0	0	0	0	0	0	6	28
	HDD65	2174	458	417	359	209	61	3	0	0	2	39	234	392
	CDD50	4809	58	52	123	243	453	668	848	867	667	503	226	101
	CDD65	1355	0	0	0	2	48	221	383	402	219	77	3	0
	CDH74	13027	0	0	3	20	334	2050	4169	4305	1724	415	7	0
	CDH80	4480	0	0	0	0	46	625	1647	1731	388	43	0	0
Wind	WSAvg	6.8	6.3	6.6	7.2	7.5	7.2	7.3	7.2	6.9	6.4	6.0	6.4	6.0
Precipitation	PrecAvg	24.0	2.4	1.9	1.4	1.9	1.5	0.6	0.2	1.0	2.8	3.2	3.9	3.3
	PrecMax	34.8	6.4	4.3	3.1	6.2	8.7	1.9	1.3	4.4	10.0	7.6	11.9	9.3
	PrecMin	12.9	0.0	0.2	0.2	0.2	0.1	0.0	0.0	0.0	0.3	0.4	0.2	0.0
	PrecStd	5.7	1.6	1.3	0.9	1.5	1.9	0.6	0.3	1.1	2.0	1.9	2.6	2.6
	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.3	67.5	71.8	76.9	84.9	94.2	97.0	96.4	90.0	84.1	74.6
MCWB			57.1	57.2	58.3	60.5	64.9	71.5	70.3	72.6	73.1	69.7	65.1	59.8
2%		DB	63.7	64.3	68.1	73.1	80.5	89.5	92.7	92.2	86.2	80.6	71.8	64.9
		MCWB	56.2	55.4	56.9	59.4	64.0	70.1	71.6	73.2	72.7	70.0	63.5	58.3
5%		DB	61.6	62.3	66.0	70.4	77.3	86.1	89.7	89.6	84.1	78.4	69.7	63.5
		MCWB	55.3	54.9	56.5	59.1	63.2	70.0	72.1	73.7	71.8	69.0	62.2	57.7
10%		DB	59.9	60.4	64.0	68.2	75.1	83.6	87.5	87.6	82.2	75.9	67.6	62.0
		MCWB	54.3	54.0	55.8	58.6	63.2	69.1	72.1	73.6	71.3	67.5	60.9	56.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.6	59.3	61.5	64.2	69.4	77.5	78.9	80.1	77.2	74.1	67.2	61.9
		MCDB	63.4	64.4	66.9	70.3	76.8	85.7	85.9	87.1	84.1	79.3	71.5	65.2
	2%	WB	58.0	57.5	59.6	62.3	67.5	74.5	77.1	78.5	75.5	71.9	65.2	60.1
		MCDB	62.0	61.8	64.7	68.8	74.8	82.8	85.0	85.9	82.7	77.6	69.8	63.4
	5%	WB	56.7	56.1	58.3	61.1	66.2	72.5	75.8	77.2	74.2	70.2	63.5	58.9
		MCDB	60.6	60.4	63.4	67.5	73.9	81.3	84.1	85.0	81.5	76.1	68.3	62.7
	10%	WB	55.1	54.7	57.0	59.9	65.0	71.0	74.7	75.9	72.6	68.7	61.7	57.4
		MCDB	59.2	59.3	62.1	66.2	72.5	79.9	83.2	84.0	80.0	74.6	66.5	61.4
Mean Daily Temperature Range	5% DB	MDBR	18.3	18.6	19.7	19.6	19.7	20.4	20.2	19.4	18.2	17.8	17.1	17.5
		MCDBR	19.9	21.5	23.2	23.5	23.0	24.3	23.8	22.4	19.9	20.0	19.2	18.8
		MCWBR	14.2	14.8	14.4	13.0	11.0	11.4	10.5	10.6	9.8	11.3	12.7	13.5
	5% WB	MCDBR	18.3	19.2	20.0	20.4	19.8	21.5	19.9	19.4	17.9	18.3	17.6	17.4
		MCWBR	14.2	14.3	13.8	12.9	11.2	11.7	11.1	11.0	10.3	11.6	12.9	13.5
Clear-Sky Solar Irradiance	taub	0.339	0.362	0.407	0.429	0.450	0.459	0.475	0.466	0.446	0.415	0.375	0.344	
	taud	2.482	2.407	2.267	2.227	2.206	2.220	2.187	2.219	2.262	2.329	2.389	2.461	
	Ebn at Noon	262	270	268	269	265	262	256	255	252	247	244	250	
	Edh at Noon	25	31	39	43	45	45	46	43	39	33	27	24	
All-Sky Solar Radiation	RadAvg	693	949	1396	1785	2072	2332	2281	1998	1492	1085	726	607	
	RadStd	42	108	113	102	120	92	86	80	80	80	57	40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+175	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page